

Perception-Driven Path Planning:

Integrating Image Segmentation and Nav2 for Autonomous Vehicles

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Motivation

- Commercial AV stacks are costly & closed
 - Proprietary sensors/software limit research access
- Need for a low-cost, modular platform
 - A system anyone can build and extend easily
- Real-time fusion of perception + planning
 - Semantic understanding directly drives navigation



Objectives



Build a bicycle-kinematic robot

Models real car-like
non-holonomic constraints

Integrate vision, LiDAR and GPS/IMU

Fuses multiple cheap sensors for
reliability

Launch:
Date



Implement waypoint navigation using Nav2

Follow GPS waypoints
autonomously

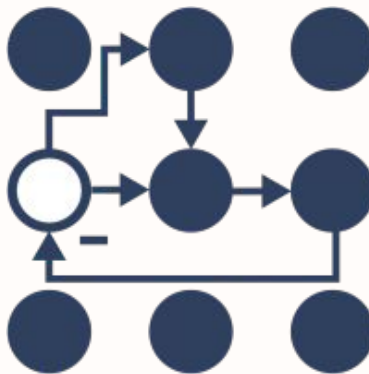
Core Technology Overview



ROS 2

Open-source robotics
middleware

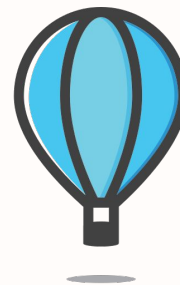
DDS messaging & lifecycle
nodes for real-time



Ros2_control

Hardware abstraction
layer

Standard joint
state/command interfaces



N A V 2

Nav2

ROS 2-native navigation
stack

Global/local planners +
layered costmaps

The Vehicles



Sensor Suite



Velodyne VLP-16 LiDAR

360° 3D scans at 10 Hz
for obstacle detection



ZED X Stereo Camera

Dual-lens depth at 15
FPS for lane &
segmentation



MTi-680G GPS/IMU

Global positioning +
orientation fusion

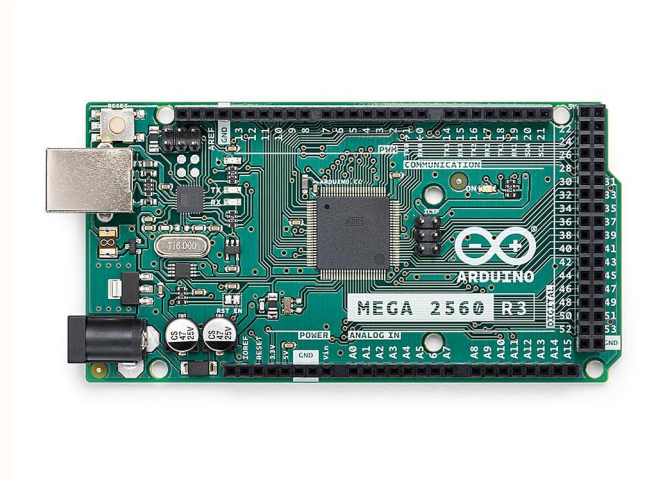


Encoders & Hall Sensor

Wheel velocity and
steering angle feedback

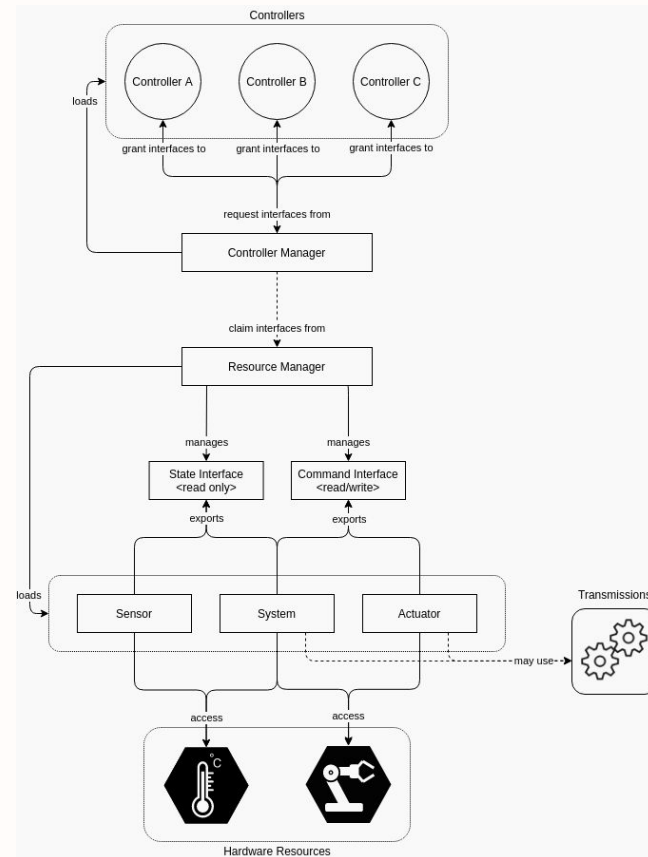
Arduino Firmware

- Dual PID loops
 - Velocity @ 30 Hz, Steering @ 3 ms for tight control
- Ziegler-Nichols initial tuning
 - Empirically refined during field tests
- PWM ramping & output clamping
 - Prevents motor stalls and jerky motion



Ros2 Control Architecture

- Carlike Bot System Hardware Interface
 - Implements ROS 2 control interfaces for the robot's joint model
 - Exports rear_wheel_joint (velocity) & front_steering_joint (position)
 - Matches the URDF joint names used in the robot model
 - Ties ROS 2 controllers directly to physical actuators



Motion Planning Models

REEDS_SHEPP

Forward &
reverse arcs

Handles tight turns and
backing-up maneuvers

DUBIN

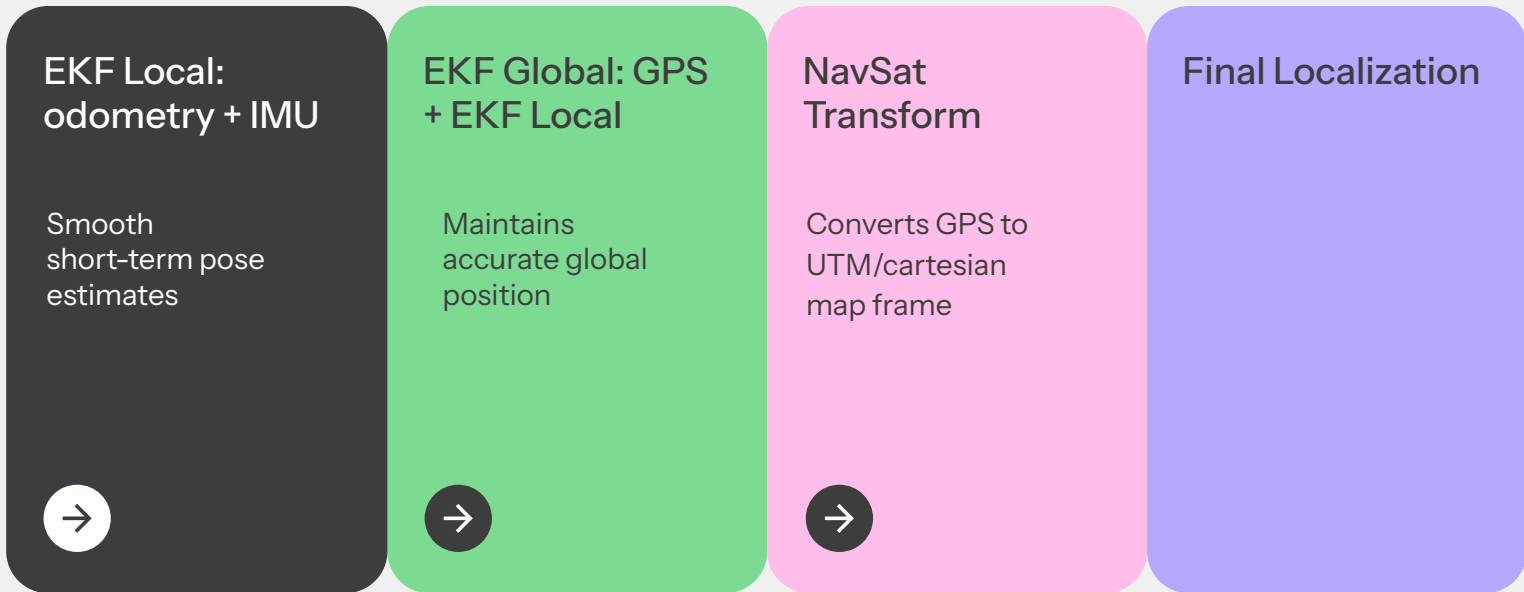
forward-only
arcs

Yields smoother, simpler
paths in open areas

Smac Planner Hybrid

Searches a lattice of these
primitives for optimal paths

Localization Pipeline



Perception Pipeline

YOLOPv2 segmentation

Real-time
drivable area &
obstacle masks

Stereo depth fusion

Places masks
accurately in
3D space

LiDAR → 2D scan

Dynamic
obstacles feed
directly into
costmap

Gazebo Teleoperation Simulation

- Ackermann-steering model in Gazebo
 - Simulated full URDF with `ros2_control` interfaces under realistic vehicle kinematics
- ROS 2 keyboard teleoperation
 - Used `teleop_twist_keyboard` to send velocity and steering commands via arrow keys
- Gazebo + RViz side-by-side
 - Validated physical motion in Gazebo while observing costmaps and TF in RViz
- Bicycle-kinematic on hardware
 - Switched to our bicycle-style model for real-world testing

Gazebo Teleoperation Simulation

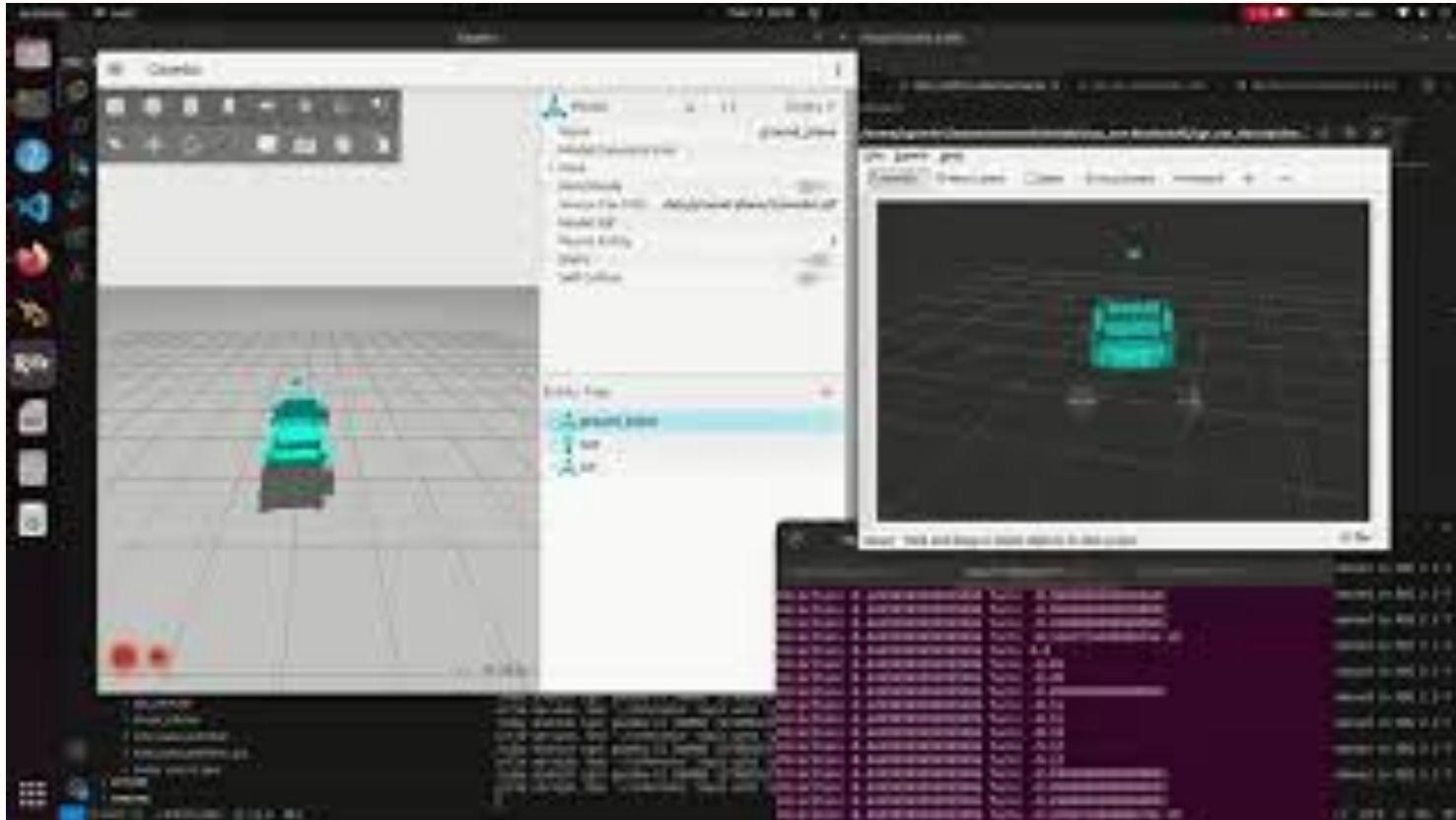


Image Segmentation Demo



3D Depth Estimation Demo

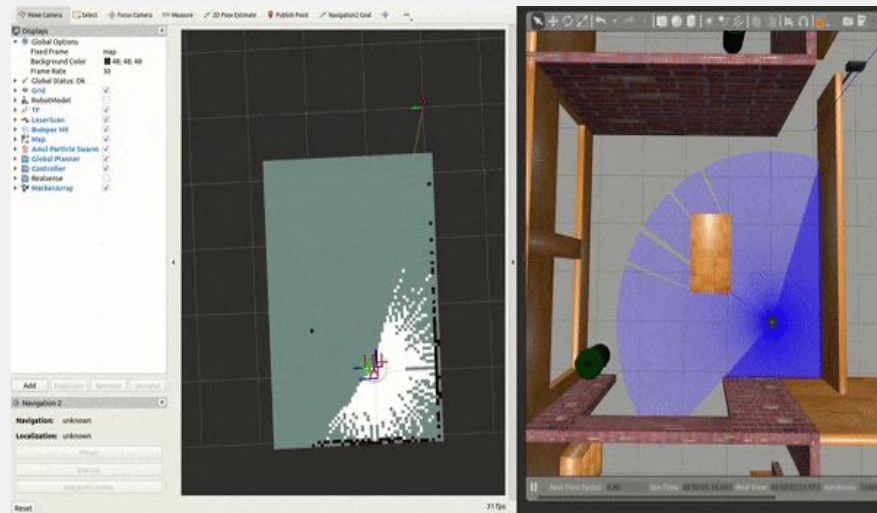


Lidar Points to 2D and Path Planning



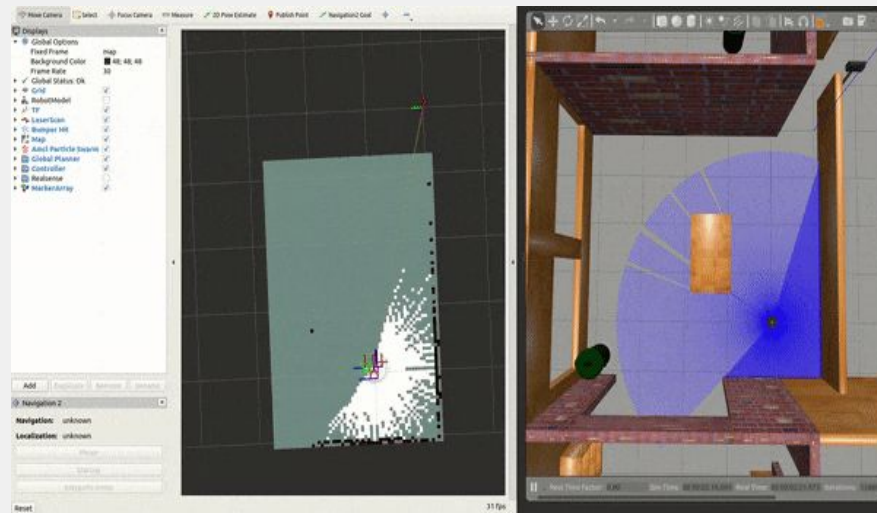
Nav2 Planner Configuration

- Smac Planner Hybrid uses motion primitives
 - Abstracts vehicle kinematics into discrete forward/back arcs
- REEDS_SHEPP vs. DUBIN models
 - REEDS_SHEPP allows both forward and reverse; DUBIN only forward
- Planner biases
 - Adjusts penalties to prefer forward motion and respect curvature limits
- Real-time planning constraints
 - Timeouts ensure the robot replans quickly in dynamic environments



Interactive Goal Setting via RViz

- Interactive goal placement
 - Use RViz “2D Nav Goal” tool to click destinations on the map
- PoseStamped from RViz markers
 - RViz converts clicks into ROS PoseStamped messages
- Real-time execution
 - Nav2 immediately plans and follows each clicked goal
- Demo-ready simplicity
 - Just point and click in RViz



Autonomous Driving at Night Time



Results and Perception Metrics

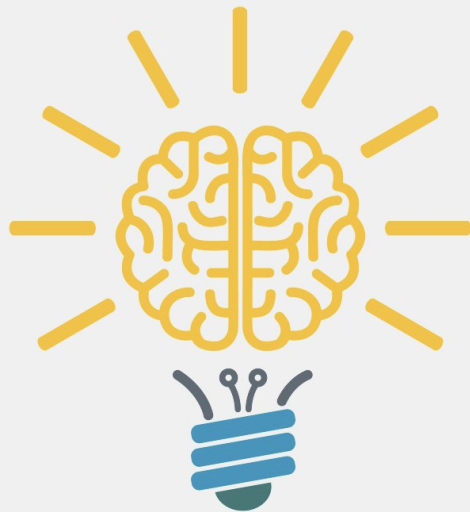
- Navigation success: 93.3 % (28/30)
 - High reliability in outdoor trials
- Avg. deviation: ± 0.35 m
 - Acceptable lateral error for low-speed autonomy
- Avg. completion: 42.8 s/10 m
 - Consistent performance across missions
- Lane Detection Accuracy: 91.2 %
 - High segmentation accuracy on varied road marks
- Object Detection : 86.5 %
 - Reliable detection at up to 18 m range
- Inference speed: 13–15 FPS
 - Real-time on Jetson Orin Nano

Challenges

- Hardware driver failures
 - Intermittent USB/Ethernet disconnects required robust reconnection logic
- Steering mount mechanical failure
 - Repeated stress on the linkage led to deformation, necessitating a stiffer bracket
- Sensor calibration drift
 - Hall sensors and encoders exhibited non-linear errors over time without regular recalibration
- Noisy odometry readings
 - Wheel slip and terrain irregularities introduced spikes in velocity estimates, IMU data was noisy.

Lessons Learned

- Modular ROS 2 design
 - Eases integration and debugging
- Extensive visualization (RViz)
 - Critical for system validation
- Precision sensor calibration
 - Underpins reliable perception and control



Future Work

- Integrate semantic masks into Nav2 costmaps
 - Enable true perception-driven obstacle avoidance
- Implement SLAM for GPS-denied areas
 - Hybrid GPS/vision localization fallback
- Behavior trees for high-level logic
 - Stop-and-wait, replan, failure recovery
- Web/mobile dashboard
 - Live mission upload, monitoring, & diagnostics





Thank you!